Pediatric drug reaction with eosinophilia and systemic symptoms: A 12-year retrospective study in a tertiary center.

PMID: 38214543 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome is a rare and severe adverse drug reaction involving multiple organs. Data on DRESS syndrome among children are currently limited. The purpose of this study was to determine the clinical features, causative drugs, systemic organ involvement, laboratory findings, disease severity score, and treatment outcomes in pediatric DRESS patients. The medical records of all pediatric DRESS patients, based on the RegiSCAR diagnostic criteria and admitted to King Chulalongkorn Memorial Hospital, Bangkok, Thailand from January 2010 to December 2021, were reviewed. Twenty-two cases were identified (males 54.5%) with a median age of 9.5 years. Anticonvulsants (54.5%) and antibiotics (27.3%) were the leading culprit drugs. Skin rash was reported in all cases, followed closely by liver involvement (95.5%). Eosinophilia and atypical lymphocytosis were identified in 54.5% and 31.8% of cases, respectively. The median latency period was 17.5 days. Liver enzyme elevation was detected at an average onset of 20.0 days and hepatocellular type was the most common pattern of liver injury. Nineteen patients (86.4%) were treated with systemic corticosteroids with prednisolone being the most prescribed medication. One case developed Graves' disease after DRESS and multiple relapses of DRESS. One case (4.5%) died due to refractory status epilepticus that was unrelated to DRESS. Anticonvulsants were the major cause of DRESS in pediatric patients. High suspicion for DRESS is crucial in patients receiving these drugs and presenting with fever, rash, and internal organ involvement.

Acute kidney injury in lamotrigine-induced DRESS syndrome.

PMID: 38801453 | Indexed in PubMed

We present a case of lamotrigine-triggered DRESS (drug reaction with eosinophilia and systemic symptoms) syndrome with acute kidney injury stage 3. A 17-year-old girl with known epilepsy treated with lamotrigine presented with acute kidney injury as well as skin eruption, fever, and apathy. Extended diagnostics, considering infectious and autoimmune diseases, remained unremarkable. Lamotrigine blood levels were within the target range. Kidney biopsy showed acute interstitial nephritis with tubular necrosis. Methylprednisolone pulse therapy led to an improvement in kidney function; skin eruption and neurological symptoms resolved. During the hospital stay, the girl admitted to inconsistent and variable intake of lamotrigine, occasionally resulting in notable overdosing. This report demonstrates that acute kidney injury in lamotrigine-induced DRESS syndrome is an acute interstitial nephritis with tubular necrosis, an aspect that has not been deeply characterized so far. Additionally, we aim to elevate awareness towards non-adherence as cause of disease, especially among the adolescent population.

How to identify DRESS, drug reaction with eosinophilia and systemic symptoms?

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DRESS (drug reaction with eosinophilia and systemic symptoms) is a rare, severe multiorgan adverse drug reaction. Antiepileptic age's and antibiotics are the most frequently reported causative agents. Compared with other drug reactions, DRESS demonstrates a long latency period thus complicating recognition and diagnosis. DRESS is defined as presence of fever, skin eruption, hematologic abnormalities and systemic involvement, especially liver injury. Withdrawal of the culprit drug, commencement of systemic corticosteroid and supportive care are the mainstay of treatment. The majority of patients recover completely after drug withdrawal and appropriate therapy. Some patients suffer from chronic sequelae or even death.

Drug reaction with eosinophilia and systemic symptoms (DRESS syndrome).

PMID: 27310545 | Indexed in PubMed

To review the hypersensitivity reaction to drugs known as drug reaction with eosinophilia and systemic symptoms (DRESS syndrome), based on a case report. We also intend to discuss the difficulty and importance of disease recognition, since none of the changes is pathognomonic of this disease and failure to identify it may have disastrous consequences for the patient. To describe this case report, in addition to the information collected for clinical assessment, a literature review was performed in the PubMed and Bireme databases in order to retrieve the latest information published in literature on DRESS syndrome. The case of a 20-year old patient is reported. After anamnesis, physical examination and laboratory tests a diagnosis of DRESS syndrome was performed, characterized by rash, hematologic alterations, lymphadenopathy and lesions in target organ. This is a rare syndrome, whose frequency varies according to the drug used and the immune status of the patient, being more often associated with the use of anticonvulsants. The approach and discussion of the topic are of paramount importance, in view of the potential lethality of this treatable syndrome. Recognizing the occurrence of DRESS syndrome and starting treatment as soon as possible is crucial to reduce the risk of mortality and improve prognosis.

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) Syndrome and the Rheumatologist.

PMID: 28138822 | Indexed in PubMed

The purpose of the review is to summarise the various drugs used in rheumatology practice implicated in the causation of DRESS syndrome. The most commonly reported drugs are allopurinol, sulfasalazine and minocycline, which pose a very high risk for DRESS syndrome development, followed by strontium ranelate and dapsone. Other, less commonly reported, drugs include leflunomide, hydroxychloroquine, non-steroidal anti-inflammatory drugs, febuxostat, bosentan and solcitinib. Reaction to some drugs is strongly associated with certain HLA alleles, which may be used to screen patients at risk of serious toxicity. DRESS syndrome is a serious reaction to many drugs used in rheumatic diseases, with a potentially fatal outcome and needs to be considered in any patient started on these medications who presents with a rash, fever and eosinophilia, sometimes with internal organ involvement.

Death Due to Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) Syndrome: A Case Report.

PMID: 37424979 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome is a life-threatening, multi-organ adverse drug reaction with an incidence of 1 in 1000 to 1 in 10 000 high-risk drug exposures. An elderly female presented to the hospital with progressive weakness and a diffuse erythematous macular rash covering most of her body that started 3 days prior. Over the next 3 days, the patient quickly deteriorated, developing disorientation with acute onset left-sided weakness, leukocytosis, thrombocytopenia, eosinophilia, liver and kidney failure, and hypoxia. Clinical and histological changes supported the diagnosis of DRESS syndrome caused by intravenous (IV) ampicillin administered during a prior hospitalization for a urinary tract infection. Systemic corticosteroids were initiated quickly thereafter, but the patient ultimately succumbed to complications caused by DRESS syndrome. There are currently no randomized trials evaluating treatments for DRESS, and evidenced-based guidelines are lacking. Viral reactivation has also been suggested as a possible complication of DRESS syndrome, though the true incidence and association remain unclear. Although we started our patient on high-dose IV corticosteroids early in her course, she still succumbed to complications of DRESS syndrome. Further research into the treatment of DRESS syndrome and its association with viral reactivation is essential.

Anti-tubercular therapy causing Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) syndrome.

PMID: 33469620 | Indexed in PubMed

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) syndrome is an idiosyncratic drug reaction following a characteristic long latency period. It is previously known as drug induced delayed multiorgan hypersensitivity syndrome (DIDMOHS) or drug induced hypersensitivity (DIHS). The syndrome is manifested by wide range of clinical symptomatology that hold a potential to be life threatening but still is under recognised. The major drugs that cause DRESS syndrome are anticonvulsants, followed by sulfonamides and many anti-in˜ ammatory drugs.Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) syndrome is an idiosyncratic drug reaction following a characteristic long latency period. It is previously known as drug induced delayed multiorgan hypersensitivity syndrome (DIDMOHS) or drug induced hypersensitivity (DIHS). The syndrome is manifested by wide range of clinical symptomatology that hold a potential to be life threatening but still is under recognised. The major drugs that cause DRESS syndrome are anticonvulsants, followed by sulfonamides and many anti-inflammatory drugs.

[Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS): a review].

PMID: 23954978 | Indexed in PubMed

DRESS (Drug Reaction with Eosinophilia and Systemic Symptoms), also known as hypersensitivity syndrome (HSS), is a severe and potentially life-threatening drug reaction. The worldwide rate of mortality is about 10 %, its incidence is increasing. Drugs that most frequently induce DRESS are aromatic anticonvulsive drugs (carbamazepine, lamotrigine, phenobarbital), and more recently new retroviral therapies. The pathogenesis of DRESS is not yet fully understood, but is certainly multifactorial involving a combination of immune reactions, ethnic predisposition, genetically determined enzyme deficiencies and reactivation of herpes viruses (HHV-6, HHV-7, EBV, CMV). Because of the involvement of the skin and internal organs, the clinical picture can be very variable. No specific clinical, histologic or laboratory parameter is available, so the diagnosis has to rely on the clinical appearance. The long latency period between start of drug intake and the initial manifestation of DRESS and the successive onset of skin and organ involvement complicates the early diagnosis. Although or even because DRESS represents a diagnostic challenge, detailed knowledge about this disease is of utmost importance to enable early therapeutic actions.

Case Series on DRESS: An Unpredictable Adverse Drug Reaction.

PMID: 37654627 | Indexed in PubMed

Drug rash with eosinophilia and systemic symptoms syndrome (DRESS) is a potentially life-threatening, drug-induced, multi-organ system reaction. The most frequently involved organ is the liver, followed by the kidneys and lungs. Early detection and diagnosis followed by withdrawal of the offending agent is vital to minimise the associated morbidity and mortality, and a detailed drug history is vital to identify the causative drugs. Although Spanish guidelines were developed by a panel of allergy specialists from the Drug Allergy Committee of the Spanish Society of Allergy and Clinical Immunology (SEAIC) and are available in literature from 2020, many clinicians are still unaware about the management of this syndrome. Framing National guidelines for the early diagnosis and pharmaco-therapeutic management of DRESS will help healthcare professionals to save the patients from unintended vulnerability. We hereby present a case series on DRESS.

DRESS syndrome: a detailed insight.

PMID: 29519170 | Indexed in PubMed

Drug reaction with eosinophilia and systemic symptoms (DRESS) syndrome is a serious and potentially fatal adverse effect to therapeutic medications. The incidence of this condition varies among different ethnicities because of the difference in the genetic makeup. Though fever, rash and eosinophilia are essential features for the diagnosis of this syndrome, these vary from patient to patient along with the involvement of various organs such as liver, kidney, lungs, pancreas, etc. Some of the atypical features are dysphagia, agranulocytosis, and chylous ascites. Phenytoin, phenobarbitone, carbamazepine, and allopurinol are the most common drugs responsible for developing this syndrome, although the list is fairly long. Among the criteria used for the diagnosis of DRESS syndrome, European Registry of Severe Cutaneous Adverse Reactions to Drugs and Collection of Biological Samples (RegiSCAR) criteria is the most commonly used one. The management of this syndrome involves early removal of the causative agent and treatment with anti-histamines and emollients in the mild form, corticosteroids in the moderate form and plasmapheresis in the severe form along with other alternatives drugs. Healthcare professionals should be more vigilant about the early manifestations of this syndrome, as early diagnosis and treatment improve outcomes considerably.